

Orion: A Multi-Agent Framework for Optimizing RAG Systems through Specialized Agent Collaboration

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Abstract

Retrieval-Augmented Generation (RAG) systems in enterprise environments face challenges including semantic overlap, ambiguous requirements, and precise knowledge retrieval difficulties. Despite Agent-based methodological advances, limitations persist in collaborative efficiency and contextual adaptability. To address these challenges, this paper introduces the Orion framework, a multi-Agent collaborative RAG architecture for Enterprise Guidance Q&A Systems. The framework integrates three specialized Agents: Information Amplification Specialist enhances content presentation to resolve similarity-based errors; Interaction Analyst optimizes query formulation to address non-expert articulation deficiencies; and Query Complexity Evaluator selects appropriate language models based on query characteristics. Empirical evaluation demonstrates 93.43% question-answering accuracy in enterprise scenarios, significantly outperforming existing systems. This multi-Agent framework enhances knowledge quality, improves retrieval precision, and generates responses better aligned with user requirements, establishing novel pathways for enterprise knowledge management and guidance question-answering systems.

CCS Concepts

• **Computing methodologies** → *Natural language generation.*

Keywords

Retrieval-Augmented Generation, Multi-Agent Systems, Enterprise Guidance Q&A System, Large Language Models

*Both authors contributed equally to this research.



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1 Introduction

With the development of large language models (such as GPT-4 [24], PaLM[1], LLaMA[10], Deepseek[4], etc.) [21, 44], the field of natural language processing (NLP) has seen significant breakthroughs. These models, trained on vast amounts of data, possess strong capabilities in contextual understanding and language generation. However, when applied to domain-specific tasks, the lack of sufficient specialized knowledge often leads to the generation of outdated information and hallucinations[15]. To address this issue, the Retrieval-Augmented Generation (RAG)[42] system incorporates external knowledge sources, enhancing generative capabilities through retrieval mechanisms, thereby significantly improving task performance[34]. Despite the progress made by RAG systems, challenges remain. Traditional RAG systems often rely on a single retrieval mechanism and a general-purpose generative model, which may fail to fully leverage their potential when dealing with complex contextual tasks[26].

Despite recent advances in NLP tasks, RAG systems face several significant limitations[3]. First, semantic ambiguities in user queries, stemming from limited retrieval expertise, substantially impact performance, with research indicating LLMs achieve correct answer rates below 50% even with extensive document retrieval[38]. Second, conventional retrieval mechanisms based on keyword matching and basic similarity calculations fail to capture deep semantic relationships between queries and documents[43], resulting in retrieved content that misaligns with user needs and compromises output quality. Third, RAG systems often rely on external knowledge bases containing outdated or incomplete information[11],

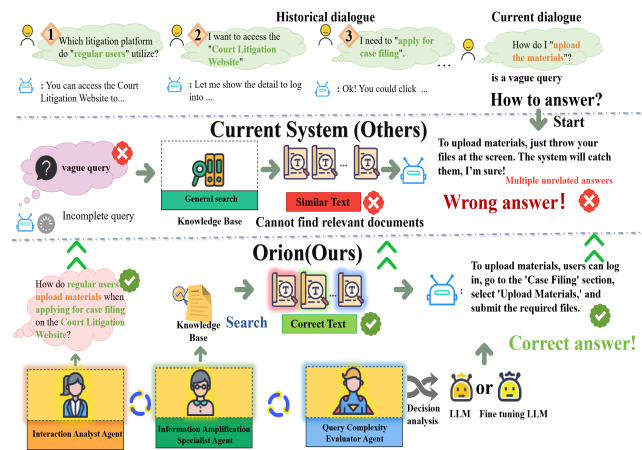


Figure 1: A RAG framework with multiple Agents, each focusing on sub-problems at every stage to solve complex tasks.

constraining their effectiveness for tasks requiring current or specialized knowledge. Finally, in local deployments without network connectivity, these systems’ dependence on pre-stored knowledge becomes more pronounced[31], particularly affecting performance in scenarios requiring precise operational information.

In response to the multiple challenges faced by traditional RAG systems, researchers have turned their attention to the integration of Agent technology with RAG. The development of Agent technology has significantly enhanced the capabilities of AI systems [31][2]. As agents responsible for executing complex tasks, LLM-enhanced Agent frameworks can autonomously generate action sequences or coordinate multiple sub-agents, thereby improving the system’s ability to handle complex scenarios [2]. Recent studies have shown that combining RAG with an Agent architecture has become an effective way to overcome the limitations of traditional RAG systems[31]. This integrated approach divides RAG into multiple specialized agents based on function, with each agent focusing on a specific task or role, thereby enhancing the system’s ability to handle diverse tasks and improving execution efficiency. Agent-based RAG systems, through the collaborative work of multiple agents, significantly enhance the overall performance of information retrieval and generation, particularly demonstrating clear advantages in handling complex domain-specific tasks[31].

Agent-based RAG systems, despite their advantages, face challenges in practical environments, including software diversity, management complexity, and LLM integration issues[30]. In scenarios with high semantic overlap, ambiguous queries, and difficult knowledge retrieval, these systems struggle to maintain answer quality, satisfy personalized needs, and process vague requests without individual context understanding[8]. These limitations frequently prevent industrial implementation despite growing academic interest. In company guideline systems where RAG uses operation manuals as knowledge bases, three semantic confusion types typically emerge: similar terms having different meanings across platforms; operational steps varying slightly between user roles on the same platform; and user expression ambiguity causing misinterpretation.

Based on this practical scenario¹, this study introduces the Orion framework, which centers on LLMs with task-specific modules and creates three complementary professional Agents.

- Information Amplification Specialist Agent, enhances the presentation of information and addresses answer errors caused by high textual similarity within the knowledge base;
- Interaction Analyst Agent, focused on understanding and optimizing user queries, solving issues of information loss and errors caused by unprofessional expressions due to lack of user experience;
- Query Complexity Evaluator Agent, which intelligently and dynamically selects the appropriate LLM based on query complexity to improve answer generation quality and accuracy.

These three Agents work in collaboration, As shown in Figure 1 forming a complete RAG application process encompassing data processing, query construction, and answer generation. To verify the effectiveness of the Orion framework, we constructed a dedicated test dataset based on the actual application scenario of a company’s internal guideline system, and conducted a comprehensive evaluation from two dimensions: answer accuracy and answer generation quality. The experimental results show that the Orion framework achieved a 93.43% answer accuracy rate in real-world application scenarios, significantly outperforming current mainstream RAG systems, confirming the effectiveness and practical value of the proposed approach. The multi-agent design of the Orion framework, particularly the collaboration of the three professional Agents, effectively addresses the diverse user interaction environment and complex system management needs. These Agents optimize document retrieval, user query understanding, and answer generation quality in various task scenarios, ensuring high efficiency and accuracy of the question-answering system by dynamically selecting the appropriate LLM and enhancing information presentation capabilities.

The contributions of this study can be summarized as follows: (1) We propose Orion, a multi-Agent framework with three specialized collaborators (Information Amplification Specialist, Interaction Analyst, Query Complexity Evaluator) that enhances traditional Agent-based RAG in semantic overlap and ambiguous query scenarios. (2) We develop intelligent information processing where Information Amplification Specialist improve knowledge base presentation quality while Interaction Analyst optimize user query interpretation, compensating for non-professional expressions in complex contexts. (3) We design an intelligent model matching mechanism where the Query Complexity Evaluator analyzes query characteristics and selects the most appropriate LLM based on different models’ capabilities rather than resource considerations, improving answer quality across diverse query scenarios.

¹The framework of this study is based on a sensitive project, the data and specific content of which cannot be publicly disclosed due to privacy protection and security requirements. Nevertheless, the design and implementation of this framework are adaptable to similar environments and data. Therefore, any application that meets the characteristics of such environments and data can adopt this framework.

2 Related work

2.1 Retrieval-Augmented Generation (RAG) Systems

Retrieval-Augmented Generation (RAG) systems combine large-scale pre-trained language models with external knowledge sources, effectively addressing issues such as outdated knowledge and generation hallucinations. The RAG framework, first proposed by Lewis et al. (2021)[19], significantly improved the accuracy of generated results. However, traditional frameworks rely on general retrieval mechanisms, which exhibit limitations in domain-specific applications. To address this issue, recent research has focused on optimizing the synergy between retrieval and generation processes. For example, the FiD model [14] enhances content consistency by processing multiple retrieved documents in parallel; UltraRAG[20, 33, 37, 39, 46], introduces a dynamic retrieval-generation iterative mechanism, which updates the retrieval strategy in real-time based on generated content. Additionally, the GraphRAG (Microsoft, 2024)[13] leverages knowledge graphs to enhance semantic connections, effectively supporting multi-hop reasoning; the HippoRAG (Berkeley, 2024)[12], designed with a dual-layer memory architecture, optimizes knowledge integration and forgetting mechanisms; and the IM-RAG[36], which supports multi-turn retrieval interactions through internal monologue learning, significantly alleviates the conflict between retrieval granularity and generation consistency, enabling dynamic knowledge updates, enhancing multi-hop reasoning, and improving the interactive experience.

At the same time, agent-based frameworks have introduced a new paradigm of modularity and collaboration for NLP tasks, drawing widespread attention from the research community. For instance, the single-agent framework Toolformer (Schick et al., 2023)[28] expands tool interfaces, enabling LLMs to autonomously call APIs to complete complex tasks. Meanwhile, the development of multi-agent systems has further promoted task specialization and collaborative efficiency. For example, the LLM-Agent Society (Liu et al., 2023)[40] constructs a collaborative framework composed of specialized agents that can optimize generation quality through structured communication protocols. To overcome the inherent limitations of RAG systems, researchers have begun to introduce agents, which have shown significant advantages. These systems are capable of flexibly handling multiple knowledge sources and multi-step tasks. For example, CRAG (Chen et al., 2024)[25] employs self-checking agents to filter out irrelevant content, while M-RAG (Multiple Partition RAG)[35] partitions databases into multiple segments, each acting as a basic unit for RAG execution. By leveraging multi-agent reinforcement learning, M-RAG optimizes various language generation tasks, significantly enhancing the system's reliability and adaptability.

2.2 Application of Agentic RAG

RAG technology has demonstrated significant application potential across various fields. In text summarization tasks, RAG systems can generate more accurate and comprehensive content; in code generation, RAG improves the practicality of the output code; in healthcare, RAG provides more precise diagnostic and treatment recommendations; and in the financial and educational sectors, RAG generates

more reliable market insights and personalized learning content. In interactive question-answering tasks, Kurnia Muludi et al.[22] validated the competitive advantages of RAG. Wen Zhang et al.[41] proposed a unified and trustworthy RAG framework that supports multiple structured data representations, improving answer accuracy through conditional graphs, two-stage queries, and dynamic sample retrieval. Jiejun Tan et al.[32] innovatively introduced an HTML-based retrieval-augmented generation paradigm, enhancing the input quality of RAG systems through HTML optimization strategies.

Researchers have proposed various specialized solutions to address the unique requirements of enterprise internal guideline systems. Parth Sarthi et al. (2023)[27] innovatively applied recursive embedding techniques in combination with a bottom-up tree structure organization method, significantly improving the accuracy of document retrieval. Masoomali Fatehkia et al. (2024)[7] took an organizational structure approach, using a tree structure to precisely represent the hierarchical relationships of internal entities within enterprises, while also introducing locally deployed open-source model architectures to address data privacy concerns. Guanting Dong et al. (2023)[6] significantly enhanced the application performance of RAG systems in enterprise knowledge management through a systematic preference knowledge construction process and five novel query enhancement strategies. Although these methods have yielded notable results in specific scenarios, there are still numerous challenges in real-world enterprise applications, such as unclear user intent expression, low collaboration efficiency between system components, and insufficient real-time adaptation to complex queries. The Orion framework proposed in this study addresses these core challenges by utilizing a collaborative mechanism involving three complementary professional agents, successfully overcoming the inherent limitations of traditional RAG systems in handling domain-specific knowledge and understanding complex contexts, thereby providing a more precise and efficient intelligent question-answering solution for enterprise internal guideline systems.

3 Methodology

The Orion framework combines the advantages of RAG architecture, effectively optimizing query and answer generation processes through intelligent multi-stage processing and Agent collaboration to ensure efficient and precise question-answering outputs. Specifically, it includes the following three key stages: **1.Information Preprocessing Stage:** The Information Amplification Specialist Agent first processes knowledge through hierarchical chunking, amplifies contextual differences in each text block, and highlights key information to prepare for subsequent query construction and answer generation. **2.Query Construction Stage:** The Interaction Analyst Agent is responsible for analyzing and optimizing user queries. This stage primarily addresses query issues arising from unclear user expressions or lack of professional knowledge. **3.Answer Generation Stage:** Based on the optimized query, the Query Complexity Evaluator Agent selects an appropriate LLM to generate the final answer according to the complexity of the query. Standard questions are processed using the base model, while non-standard questions are handled by advanced fine-tuned models,

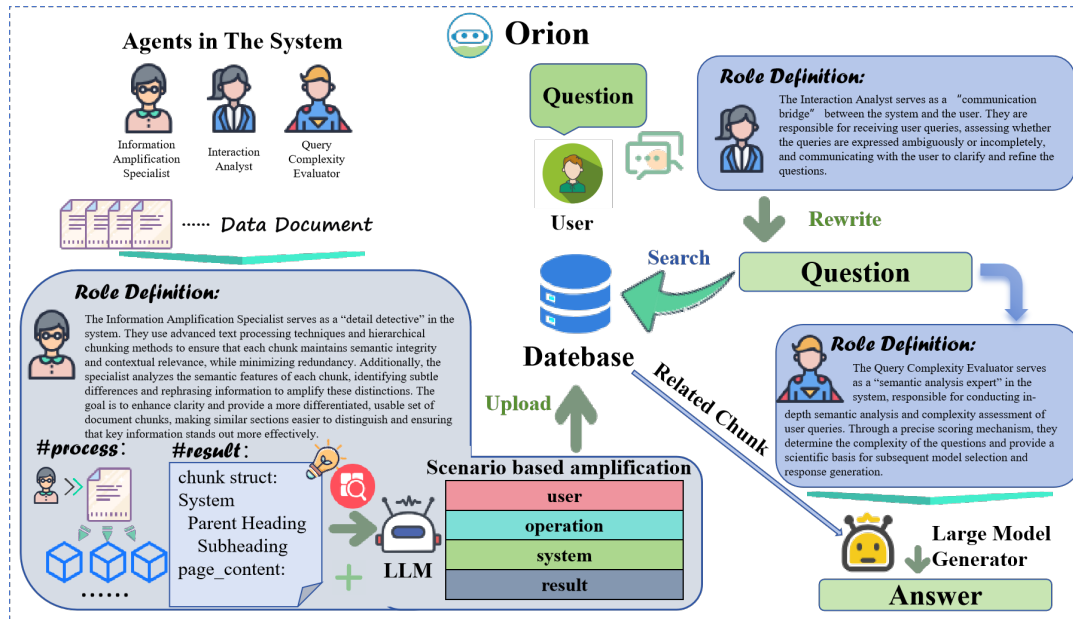


Figure 2: Orion is a multi-Agent specialized collaboration framework that designs three functionally complementary professional Agents (Information Amplification Specialist, Interaction Analyst, and Query Complexity Evaluator) to work in synergy, optimizing the limitations of traditional Agentic RAG in scenarios involving semantic overlap and ambiguous user needs.

thereby ensuring that the generated answers are both accurate and high-quality.

These three specialized Agents—Information Amplification Specialist, Interaction Analyst, and Query Complexity Evaluator—work collaboratively within the Orion framework, complementing each other to complete the entire question-answering process. Each role is optimized to address specific challenges, forming effective coordination between them. The implementation process of the Orion framework is illustrated in Figure 2 below.

3.1 Information Amplification Specialist Agent

The Information Amplification Specialist acts as the 'detail detective' in the system. It employs advanced text processing techniques and hierarchical segmentation methods to ensure that each segment maintains semantic integrity and contextual relevance, while minimizing redundancy. Additionally, it analyzes the semantic features of each segment, identifies subtle differences, and enhances these distinctions through rephrasing. The goal of this agent is to improve the data quality of the knowledge base by providing more differentiated and usable document fragments, making similar sections easier to distinguish, and ensuring that key information is highlighted more effectively.

The specific process of text segmentation begins by precisely decomposing complex, unstructured documents into easily manageable hierarchical blocks. We found that the internal guideline documents in the company typically follow strict formatting standards. Therefore, we use regular expressions to accurately identify and extract directory information from the documents, including titles and hierarchical relationships. Based on this, we construct a hierarchical tree structure, dividing the document into chapters,

sections, subsections, and other levels. Next, we further refine and consolidate the document content into individual blocks based on the smallest subheading at each level. Each block is tagged with the label of its corresponding level, serving as contextual metadata, thereby enhancing the accuracy of the retrieval process. Through this segmentation method, we effectively reduce overlap and redundancy between different blocks, while ensuring traceability within the hierarchical structure, thus improving the accuracy of both the retrieval and generation processes.

Performing differentiated amplification based on text segmentation is essential for improving system accuracy. Using the operational manuals of court internal guidance systems as an example, we identified high-similarity confusion scenarios: first, similar but distinct operation instructions across platforms (e.g., "self-litigation case filing" versus "litigation case filing"), and second, operation steps that are similar yet differ in detail for different user roles (e.g., judges establishing cases versus ordinary users applying for cases). This semantic similarity interferes with retrieval results, reducing question-answering accuracy. To address this, we designed a differentiation amplification template that generates diverse semantic expressions and enhances contextual information, reducing semantic overlap between similar text blocks. Analysis of user queries revealed consistent patterns, leading to a representational method that abstracts relationships between four core elements: the user role classification precisely defines the operation subject; the operational platform represents the action's carrier and environmental constraints; the actions the user must perform and their corresponding functions are mapped to decompose the operational process, amplifying operation details. This method structurally expresses complex operational processes when users pose questions, forming

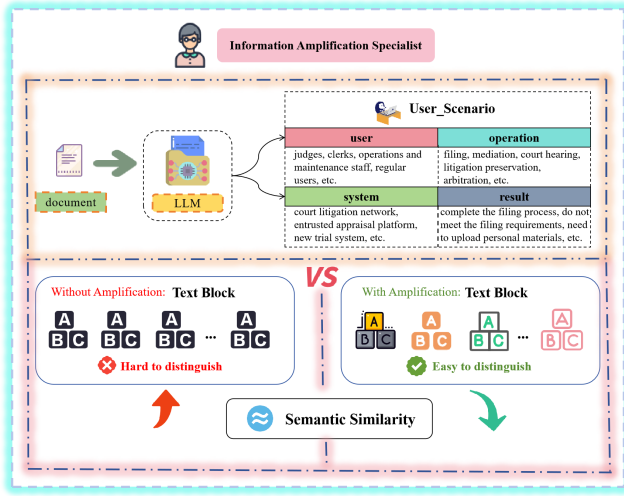


Figure 3: Workflow of Information Amplification Specialization Agent.

the basis for our unified differentiation scenario formula:

$$\text{User_Scenario} = U + S + O + R \quad (1)$$

Here, U represents the user role, S represents the platform in use, O represents the action the user needs to perform (e.g., dragging a file), and R represents the functionality corresponding to the action (e.g., file upload). Through the differentiated context formula, we can generate unique contextual descriptions for each text block, effectively reducing semantic overlap and highlighting key differences. This process provides a clearer and more precise semantic background for subsequent query construction and answer generation. Based on this context formula, we employ the large language model Deepseek-R1-Distill-Qwen-7b to capture subtle differences between text blocks and generate differentiated contexts. The detailed template can be found in Appendix A, and the specific process is shown in Figure 3. Through contextual differentiation, the semantics of document blocks become richer and more diverse. By explicitly extracting and amplifying subtle differences between similar document blocks, semantic overlap is reduced. While maintaining the core semantics of the document blocks, this approach effectively minimizes the semantic overlap between similar blocks.

3.2 Interaction Analyst Optimizing Agent

The Interaction Analyst performs rule-based validation of the user's query, integrates the user's historical query information, and infers and completes any missing content to generate a complete and accurate query. This process enhances the system's retrieval and answering performance. The specific process is illustrated in Figure 4.

When users interact with the system, due to their lack of experience or cognitive biases, or because their questions are not coherent, the user queries captured by the system are often incomplete or lack contextual continuity. This results in the system being unable to understand the user's true intentions, posing challenges for both the retrieval and generation modules. To address this issue, we

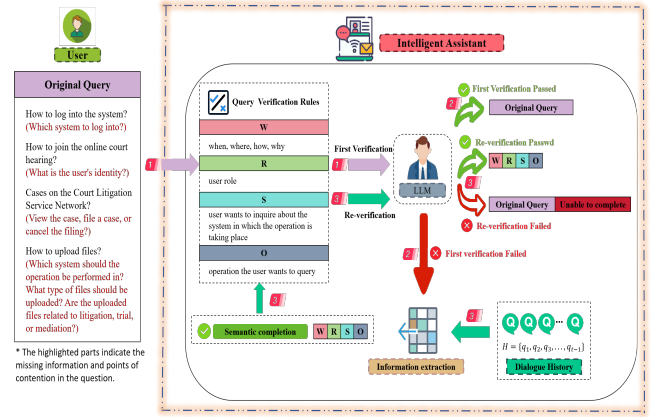


Figure 4: Workflow of Interaction Analyst Optimizing Agent.

abstract the user's query intentions based on their habitual questioning patterns in practical applications, using four key elements: the user's intention, the user, the platform, and the user's current action and completed function. For example, in the question 'Where can I upload a file to the cloud on a platform?', 'Where' represents the user's intention, 'I' represents the user, 'the platform' refers to the corresponding operation platform, and 'uploading a file to the cloud' refers to the user's current action. Based on this, we have designed a question rule validation template Q :

$$Q = W + R + S + O \quad (2)$$

Where W includes the question elements of When, Where, How, and Why, meaning any one of these question aspects must be satisfied. R represents the user's role, S represents the platform the user is currently on, and O refers to the action the user is currently about to perform.

We use a powerful language model (LLM) combined with the question rule validation template Q to determine whether there is missing information in the user's query. If the missing elements are identified as R , S , or O , historical dialogue is introduced to supplement the missing information, with the concatenation process is as follows:

$$Q_t = \text{Concat}(H, q_t) \quad (3)$$

In this context, $H = \{q_1, q_2, q_3, \dots, q_{t-1}\}$ represents the user's query history from the previous $t - 1$ turns, with q_t denoting the current query. *Concat* refers to the concatenation of the user's query history with the current question, thereby forming a complete dialogue history Q_t . After multiple rounds of conversation, considering the exponential growth of input tokens, a conversation summary is conducted after M rounds of dialogue to reduce the consumption of input tokens. For example, if the user previously asked "How to enter the case-filing interface?", and currently asks "What materials are needed?", the completion would be: "Based on the steps of 'entering the case-filing interface', the user wants to know what materials need to be prepared." In addition, if it is determined that W is missing, the LLM (Large Language Model)

combined with one-shot learning is used to identify the user’s intention and complete any of the elements of When, Where, How, or Why.

Through the above operations, the Interaction Analyst refines the semantic information of the user’s query, captures the user’s intent, and reconstructs the question. The specific prompt design is shown in Appendix B.1. In the internal court Q&A scenario, this approach effectively addresses retrieval bias caused by incomplete queries, providing users with more accurate and thoughtful responses, significantly enhancing both the precision of retrieval and the coherence of generation.

3.3 Query Complexity Evaluator Agent

The Query Complexity Evaluator, as the ‘analytical decision-maker’ in the system, is responsible for performing a deep evaluation of the user’s input query to assess its complexity. Their task is to analyze the query’s semantics, structure, and contextual information in order to provide an accurate complexity score, which informs the decision-making process regarding the selection of the appropriate generation model. The overall process is illustrated in Figure 5. During the implementation process, we first established an evaluation standard to assess whether the query aligns with the expected semantic structure, length range, and adheres to the operational or questioning standards of the specific domain, ensuring the query is clear and complete. The scoring formula is designed as follows:

$$R(q) = \text{Sim}(q, T) + \text{Len}(q) + \text{Struct}(q) \quad (4)$$

Where $\text{Sim}(q, T)$ represents the semantic similarity between the query and the question rule. $\text{Len}(q)$ denotes whether the query length falls within the expected range. $\text{Struct}(q)$ indicates whether the query structure conforms to the domain-specific standards.

This design was inspired by our observation that when faced with queries from users with varying levels of experience, inexperienced users tend to ask disorganized questions with vague wording, while experienced users often pose more complex questions based on their own reasoning. In both cases, fine-tuned LLMs tend to generate more precise responses compared to the original LLM (here, we use Deepseek-R1-Distill-Qwen-7b as the generator). These queries are categorized as deviating from the standard. For queries that follow the standard, our practical application revealed that the fine-tuned Deepseek-R1-Distill-Qwen-7b tends to

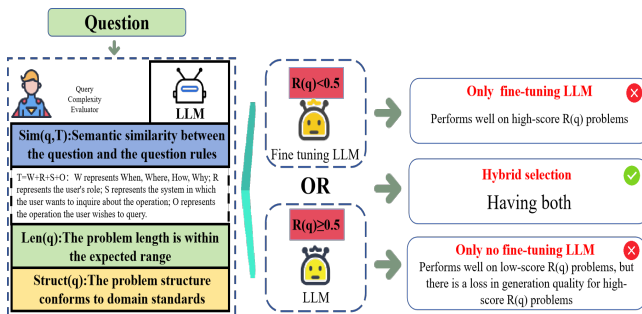


Figure 5: Workflow of Query Complexity Evaluator Agent.

generate precise and concise answers, sacrificing information completeness. In contrast, the original Deepseek-R1-Distill-Qwen-7b better demonstrates its information integration capabilities, providing detailed and complete responses. Based on this finding, we propose the aforementioned standards and incorporate them into the prompt, utilizing the large language model to provide a comprehensive score for the query on a scale from 0 to 1. The specific prompt design is shown in Appendix C.1.

Based on the $R(q)$ score provided by the large language model, an appropriate generator is dynamically selected: for $R(q) \geq 0.5$ (more aligned with the normative structure), the original Deepseek-R1-Distill-Qwen-7b is directly invoked, fully leveraging its general language capabilities and open-ended generation potential. For $R(q) < 0.5$ (deviating from the norm), the fine-tuned Deepseek-R1-Distill-Qwen-7b is used, capitalizing on its precision in specific domains. In our application, the design and selection of the generator directly influence the quality of the generated responses and the overall user experience.

Through the analytical decision-making capability of the Query Complexity Evaluator Agent, we have implemented a dynamic allocation mechanism for the generators, ensuring high-quality generation while maintaining the model’s general capabilities. This approach enhances the overall robustness of the system and the professionalism of the outputs.

In this agent, we fine-tuned Deepseek-R1-Distill-Qwen-7b using LoRA to better adapt it to the legal operation manual question-answering task, significantly reducing the risk of hallucinated generation. The specific process is as Algorithm 1 follows:

Algorithm 1 Fine-Tune Procedure for QA Generation Model

Require: Document blocks D , Number of QA pairs N , Deepseek-R1-Distill-Qwen-7b M

Ensure: Fine-tuned model M'

- 1: Initialize QA pair set $Q \leftarrow \emptyset$
- 2: **for** each document block $d \in D$ **do**
- 3: Generate N QA pairs: $q \leftarrow \text{generate_qa_pairs}(d, N)$
- 4: Append to set: $Q \leftarrow Q \cup q$
- 5: **end for**
- 6: Filter valid pairs: $Q' \leftarrow \text{filter_qa_pairs}(Q)$
- 7: Compute invalid pairs: $\text{invalid_qa_pairs} \leftarrow Q \setminus Q'$
- 8: Print removed count: $\text{len}(\text{invalid_qa_pairs})$
- 9: Truncate valid pairs: $Q_{\text{final}} \leftarrow Q'[1 : 17126]$
- 10: $\text{load_model}(M)$
- 11: Fine-tune with LoRA: $M' \leftarrow \text{fine_tune_with_lora}(M, Q_{\text{final}})$
- 12: **return** M'

4 Performance Evaluation

To validate the superiority and effectiveness of the proposed framework, we conducted multiple experimental trials using a self-constructed dataset. The experiments were divided into two scenarios: 4.3 Comparing the accuracy of our framework with that of current state-of-the-art (SOTA) frameworks and methods, and 4.4 Ablation experiments, which examine the effectiveness of each component of our method. The results demonstrate that our approach is highly effective and consistently ensures outstanding performance.

4.1 Dataset

Dataset: In order to verify the applicability and effectiveness of our proposed framework in practical application scenarios, we have designed and constructed a high-quality dataset (*MSQA_data*) containing 18,237 data items. Among them, 1,111 data items (*MSQA_test_data*) are used for performance testing, and the remaining 17,126 data items (*MSQA_train_data*) are specifically used for training the generator in the Query Complexity Evaluator Agent. The dataset originates from multiple actual operational environments and covers common queries from various user roles (such as ordinary users, judges, clerks, operation and maintenance personnel, etc.) in their work with the company's internal systems, like "How to file a case" and "How to mediate a case". Each data item includes a specific question, the correct document content for verifying the answer, and its corresponding data source, which ensures the diversity and authenticity of the test data. Through this design, we can comprehensively evaluate the performance of the framework in professional work scenarios, especially its interactive capabilities when facing users with different job responsibilities and levels of experience, as well as its adaptability in handling complex questions.

Considering that the dataset contains a vast amount of sensitive information, including the specific work content, operational processes, and project confidential information of multiple user roles, protecting these sensitive data is our primary task. Therefore, the dataset cannot be publicly shared. Despite this, the design of our framework is highly versatile and adaptable. As long as the documents comply with the hierarchical structure relationship specifications, whether they are operational manuals from other fields, knowledge base documents, or workflow documents from different scenarios, they can be seamlessly adapted and efficiently processed within our framework. This design ensures that the system is not limited to a specific dataset but can be widely applied to various documents and data sources that meet the structural requirements, thereby maintaining the system's flexibility and scalability.

4.2 Experimental Settings

Experimental Settings: In this experiment, we fine-tuned the Deepseek-R1-Distill-Qwen-7b model in the Query Complexity Evaluator using LoRA (Low-Rank Adaptation) technology. We utilized the *MSQA_train_data* dataset and set the learning rate to 5e-05, the batch size to 16, and trained for 3 epochs. During the training process, we employed the Adam optimizer and set betas=(0.9, 0.999) and epsilon=1e-08 to help avoid overfitting. Throughout the training, we regularly checked the accuracy of the validation set to ensure the stability and generalization capability of the model. In Interaction Analysis, the number of dialogue turns M for generating a dialogue summary in Q_t varies significantly across different model architectures, which directly impacts the utilization of the model's context window and computational overhead. In the context of this experimental scenario, we set M to 20, meaning that a dialogue summary is generated every 20 turns to reduce the consumption of input tokens.

Based on this, the reasoning process of the Orion framework is as follows: First, the text is chunked and differences are amplified by the Information Amplification Specialist. The processed text blocks

are then vectorized and stored in the Elasticsearch 7.5.2 server for retrieval. The user's input is restructured into a refined question by the Interaction Analyst, and the reconstructed question is used to retrieve the top K most relevant text blocks from Elasticsearch through hybrid search. We set K to 5 to balance both accuracy and reasoning speed. Subsequently, top K most relevant text blocks are input into the large model for re-ranking, and the range is narrowed down to further select the top L most relevant text blocks, where $L < K$. The Query Complexity Evaluator assigns a complexity score to the reconstructed question to select an appropriate generator. The L most relevant text blocks and the question are then input into the selected generator, which identifies the most relevant text block among the L and generates the final response based on it.

In our experiments, we used Accuracy and GPT4o-score as metrics to evaluate the correctness of answers and the quality of generation. Accuracy was used for accuracy evaluation in S1 and S2, while GPT4o-score was used for generation quality evaluation in S2.2 and S2.3. The calculation method for Accuracy is as follows: If the three text indices listed in the answer generated by the model contain the correct answer, the answer is considered correct. Traditional accuracy assessment methods typically use algorithms such as semantic similarity and keyword matching to calculate related metrics. However, in our system, due to the complexity and length of the original documents, our goal is to summarize the key steps in the text. Therefore, these conventional methods are not suitable for our purpose, we calculate accuracy based on the inclusion of the correct answer text. Additionally, our system also includes a panel that displays the top L text blocks, allowing users to quickly index and view the original document content.

$$\text{Accuracy} = \frac{1}{N} \sum_{i=1}^N \mathbb{I}(\text{Correct}_i), \quad (5)$$

$$\text{Top } L \text{ blocks}_i = \text{LLM_Rerank}(\text{Top } K \text{ blocks}_i \text{ from Retrieval System}), \quad (6)$$

$$\mathbb{I}(\text{Correct}_i) = \begin{cases} 1, & \text{if the correct citation is contained in Top } L \text{ blocks}_i, \\ 0, & \text{otherwise.} \end{cases} \quad (7)$$

Among them, N represents the total number of questions evaluated, I is an indicator function, *Top K blocks _{i} from Retrieval System* is the top K most similar text blocks retrieved from all the text blocks by vector search in the retrieval system, and *Top L blocks _{i}* is the L text blocks selected by the large model from the top K blocks through further narrowing down the range and reranking. Where $L < K$, we set K to 5 and L to 3 in this case. *Correct _{i}* is defined as follows: The retrieval system retrieves the top K most similar text blocks based on vector search and inputs them into LLM for reranking. The model then narrows down the range and selects the top L most relevant text blocks, where $L < k$. Afterward, the LLM selects the most probable text block from the top m to generate the final answer. If the selected top L blocks include the correct text block, then *Correct _{i}* = 1, otherwise *Correct _{i}* = 0.

The evaluation method for the GPT-4o score is based on a scoring template we developed to assess the quality of generated answers

Table 1: Comparison of experimental results

Method	Accuracy
R2R	0.2997
Hystack	0.5437
TrustRAG	0.6587
Ragflow	0.7281
Txtai	0.7408
FlashRAG	0.7934
Ours	0.9343

from four aspects: factual accuracy, information completeness, logical coherence, and clarity of expression.[18] For the Q&A scenarios within the internal guidance system, factual accuracy and information completeness are considered crucial in determining whether the answer effectively addresses the problem, and therefore, they are given higher weight. In contrast, logical coherence and clarity of expression are more dependent on the model’s reasoning and generation capabilities, so they are assigned lower weight. Based on this, we use GPT-4o for comprehensive evaluation to derive the final score. The detailed scoring design is provided in Appendix C.2.

4.3 MAIN RESULT(S1)

In this section, our baseline employs the current SOTA RAG frameworks, including R2R[29], Hystack[5], TrustRAG[45], Ragflow[16], Txtai[23], and FlashRAG[17].The results are shown in Table 1.

Through this evaluation method, we believe that the performance of the model in real-world scenarios can be realistically reflected.

Table 2: Comparison of experimental results of chunking methods

Method	Accuracy
Basic chunking	0.8062
Hierarchical chunking (Ours)	0.9343

Table 3: Comparison of experimental results with and without text information difference amplification

Method	Accuracy
Amplification without difference	0.8394
Amplification with difference (Ours)	0.9343

4.4 Ablation Study

In this section of the experiment, we will validate the advantages of the three Agents: (1)the role of the Information Amplification Specialist,(2)the role of the Interaction Analyst,(3)the role of the Query Complexity Evaluator.

4.4.1 The role of the Information Amplification Specialist. To validate the necessity of each component in the Information Amplification Specialist Agent, we conducted separate ablation experiments on the text fragmentation method and the difference amplification technique. In the ablation experiment for the text fragmentation method, we demonstrated the effect by comparing the impact of the basic fragmentation method and our hierarchical fragmentation method on question-answering accuracy. The basic fragmentation method divides the text into uniform-sized chunks of 500 tokens with a 15% overlap. The results of the text fragmentation ablation experiment are shown in Table 2.

As shown in Table 2, the document question-answering accuracy for the basic fragmentation strategy is 80.62%, while the accuracy for the hierarchical fragmentation strategy is 93.43%, representing an improvement of 12.81%. This demonstrates that our hierarchical fragmentation strategy effectively refines the document content, ensuring the integrity of contextual information without introducing excessive redundancy. It reduces the guidance of vague or irrelevant content, thus helping to generate more specific and relevant answers.

In text information amplification, due to the excellent inference performance of Deepseek-R1-Distill-Qwen-7b, we use it to capture the information differences between texts and effectively perform information rephrasing. We will directly conduct text retrieval and generation without applying the information amplification method to the document. A comparative experiment will be conducted with text blocks processed using the text information amplification method to examine the impact of information amplification on question answering accuracy. The comparison results are shown in Table 3.

The experimental results show that text information amplification improves question answering accuracy, particularly in documents with semantic ambiguity and overlap. Information amplification effectively helps the model distinguish details and return the most relevant content along with its corresponding document, significantly enhancing generation quality and reducing issues of vague or repetitive answers caused by low-quality data.

4.4.2 The role of the Interaction Analyst. In this section, we validate the superiority of the proposed question validation and completion template in comparison to the Explicit Instruction and Structured Context methods for user question completion tasks. We use the Prompt template shown in Appendix B.2 B.3 and conduct experimental evaluations with multiple large language models, including Chatglm4-9b[9], Qwen2.5-7b-Instruct, and Deepseek-R1-Distill-Qwen-7b. To ensure the objectivity of the evaluation, we employ GPT-4o for a multi-dimensional comprehensive quality assessment of the generated completed questions. The experimental results, presented in Table 4, provide a detailed comparison of the performance differences across various methods and models.

The experimental results show that Deepseek-R1-Distill-Qwen-7b, combined with our question completion method, achieves superior quality scores in dimension_scores, completeness, logic_rigor, and expression_clarity compared to other models and methods, and obtains a higher GPT-4o score, demonstrating excellent question reconstruction capabilities.

Table 4: Comparison of prompt effectiveness with different model combinations

Model	Prompt	dimension_scores	completeness	logic_rigor	expression_clarity	Total
Chatglm4-9b	Explicit Instruction	35	20	18	10	83
Qwen2.5-7b-Instruct	Explicit Instruction	30	20	18	9	77
Deepseek-R1-Distill- Qwen-7b	Explicit Instruction	35	25	18	9	87
Chatglm4-9b	Structured Context	35	20	18	10	83
Qwen2.5-7b-Instruct	Structured Context	30	20	16	10	76
Deepseek-R1-Distill- Qwen-7b	Structured Context	36	25	18	10	89
Chatglm4-9b	Our Approach	35	25	18	9	87
Qwen2.5-7b-Instruct	Our Approach	40	28	20	9	97
Deepseek-R1-Distill- Qwen-7b	Our Approach	40	30	20	9	99

Table 5: Comparison of experimental results with and without Interaction Analyst Agent

Method	Accuracy
Without Interaction Analyst	0.8743
With Interaction Analyst (Ours)	0.9343

Table 6: Comparison of prompt effectiveness with different model combinations

Method	Accuracy	GPT-4o score (mean)
LLama3.1-8b	0.8924	86.786319
ChatGLM-4-9B	0.9056	88.344823
Deepseek-R1-Distill-Qwen-7b	0.9103	95.655172

To further validate the importance of the entire Interaction Analyst Agent within our RAG framework, we conduct an ablation study on the Interaction Analyst Agent module to examine its impact on question answering accuracy. The comparison results are shown in Table 5.

The experimental results demonstrate that the accuracy of question answering improves after input completion by the Interaction Analyst, especially for incomplete or ambiguous questions. The Interaction Analyst helps users express their needs more clearly, reduces incorrect answers caused by unclear user inputs.

4.4.3 The role of the Query Complexity Evaluator. In this section, we first demonstrate the superiority of using Deepseek-R1-Distill-Qwen-7b as the generator. We compare multiple original LLMs with strong Chinese generation capabilities, including LLama3.1-8b, ChatGLM-4-9B, and Deepseek-R1-Distill-Qwen-7b, for generating answers to user queries. The accuracy of the answers is evaluated using Accuracy, and the quality of the generated content is assessed using GPT-4o. For the quality evaluation, we randomly select 87 samples and conduct multi-dimensional comprehensive scoring in four aspects: factual accuracy, information completeness, logical coherence, and expression clarity. The experimental results are shown in Table 6.

The experiment shows that Deepseek-R1-Distill-Qwen-7b outperforms LLama3.1-8b and ChatGLM-4-9B in both the Accuracy metric and the GPT-4o evaluation framework, demonstrating its

Table 7: Comparison of experimental results with and without fine-tuned generators, and with hybrid selection

Method	Accuracy	GPT-4o score (mean)
Generator without fine-tuning	0.9103	95.655172
Generator with fine-tuning	0.9138	95.822613
Hybrid selection (Ours)	0.9343	97.324128

advantages in content generation quality and alignment with the target content.

Next, we validate the superiority of dynamically selecting the generator method within this Agent. In this part, we conduct experiments comparing the use of an original, non-finetuned generator, a finetuned-only generator, and our dynamic cross-usage of the Deepseek-R1-Distill-Qwen-7b generator. We evaluate the accuracy of answers to user queries and the GPT-4o score as performance metrics, assessing the generated content from both precision and quality perspectives. The comparison results are shown in Table 7.

The experimental results demonstrate that the generator equipped with our dynamic cross-usage method better adapts to tasks. Whether handling complex, unstructured queries or simpler ones, our generator provides higher-quality answers with greater accuracy.

5 Conclusion

Our Orion framework, through three complementary specialized Agents, successfully addresses core challenges of traditional RAG systems in enterprise applications: data semantic overlap, ambiguous user expressions, and precise knowledge retrieval difficulties. In Enterprise Guidance Q&A scenarios, Our Information Amplification Specialist Agent resolves errors caused by similar texts; Interaction Analyst Agent optimizes query understanding to compensate for non-professional expressions; and Query Complexity Evaluator Agent matches suitable language models based on query characteristics. Experiments demonstrate 93.43% question-answering accuracy, substantially outperforming existing RAG systems.

The framework’s three key innovations—multi-agent collaboration, intelligent information processing, and model matching mechanisms—constitute an efficient enterprise-level Q&A solution that transcends the constraints of traditional Agentic RAG architectures. However, Orion still presents certain limitations. Therefore,

our future research directions include optimizing agent collaboration to enhance response latency, expanding the framework’s applicability across diverse enterprise scenarios, and incorporating user feedback mechanisms for continuous system improvement.

Overall, Orion provides an effective approach to enterprise knowledge management and intelligent Q&A challenges, offering valuable insights for industrial RAG deployment. As technology evolves, multi-Agent collaborative RAG systems will play crucial roles in broader enterprise applications.

References

- [1] Rohan Anil, Andrew M. Dai, Orhan Firat, Melvin Johnson, Dmitry Lepikhin, Alexandre Passos, Siamak Shakeri, Emanuel Taropa, Paige Bailey, Zhifeng Chen, Eric Chu, Jonathan H. Clark, et al. 2023. PaLM 2 Technical Report. arXiv:2305.10403 [cs.CL] <https://arxiv.org/abs/2305.10403>
- [2] Anthropic. 2024. *Building effective agents*. <https://www.anthropic.com/research/building-effective-agents>
- [3] Yiqun Chen, Lingyong Yan, Weiwei Sun, Xinyu Ma, Yi Zhang, Shuaiqiang Wang, Dawei Yin, Yiming Yang, and Jiaxin Mao. 2025. Improving Retrieval-Augmented Generation through Multi-Agent Reinforcement Learning. arXiv:2501.15228 [cs.CL] <https://arxiv.org/abs/2501.15228>
- [4] DeepSeek-AI, Daya Guo, Dejian Yang, Haowei Zhang, Junxiao Song, Ruoyu Zhang, Runxin Xu, Qihao Zhu, Shiroon Ma, Peiyi Wang, Xiao Bi, Xiaokang Zhang, Xingkai Yu, Yu Wu, Z. F. Wu, et al. 2025. DeepSeek-R1: Incentivizing Reasoning Capability in LLMs via Reinforcement Learning. arXiv:2501.12948 [cs.CL] <https://arxiv.org/abs/2501.12948>
- [5] deepset-ai team. 2024. *GitHub - deepset-ai/haystack: AI orchestration framework to build customizable, production-ready LLM applications. Connect components (models, vector DBs, file converters) to pipelines or agents that can interact with your data. With advanced retrieval methods, it's best suited for building RAG, question answering, semantic search or conversational agent chatbots*. <https://github.com/deepset-ai/haystack>
- [6] Guanting Dong, Yutao Zhu, Chenghao Zhang, Zechen Wang, Zhicheng Dou, and Ji-Rong Wen. 2024. Understand What LLM Needs: Dual Preference Alignment for Retrieval-Augmented Generation. arXiv:2406.18676 [cs.CL] <https://arxiv.org/abs/2406.18676>
- [7] Masoomali Fatehikia, Ji Kim Lucas, and Sanjay Chawla. 2024. T-RAG: Lessons from the LLM Trenches. arXiv:2402.07483 [cs.AI] <https://arxiv.org/abs/2402.07483>
- [8] Bijit Ghosh. 2024. *Top RAG Pain Points and Solutions*. <https://medium.com/@bijit211987/top-rag-pain-points-and-solutions-108d348b4e5d>
- [9] Team GLM, :, Aohan Zeng, Bin Xu, Bowen Wang, Chenhui Zhang, Da Yin, Dan Zhang, Diego Rojas, Guanyu Feng, Hanlin Zhao, Hanyu Lai, Hao Yu, Hongning Wang, Jiadao Sun, Jiajie Zhang, Jiale Cheng, Jiayi Gui, Jie Tang, et al. 2024. Chat-GLM: A Family of Large Language Models from GLM-130B to GLM-4 All Tools. arXiv:2406.12793 [cs.CL] <https://arxiv.org/abs/2406.12793>
- [10] Aaron Grattafiori, Abhimanyu Dubey, Abhinav Jauhri, Abhinav Pandey, Abhishek Kadian, Ahmad Al-Dahle, Aiesha Letman, Akhil Mathur, Alan Schelten, Alex Vaughan, Amy Yang, Angela Fan, et al. 2024. The Llama 3 Herd of Models. arXiv:2407.21783 [cs.AI] <https://arxiv.org/abs/2407.21783>
- [11] Zirui Guo, Lianghao Xia, Yanhua Yu, Tu Ao, and Chao Huang. 2024. LightRAG: Simple and Fast Retrieval-Augmented Generation. arXiv:2410.05779 [cs.IR] <https://arxiv.org/abs/2410.05779>
- [12] Bernal Jiménez Gutiérrez, Yiheng Shu, Yu Gu, Michihiro Yasunaga, and Yu Su. 2025. HippoRAG: Neurobiologically Inspired Long-Term Memory for Large Language Models. arXiv:2405.14831 [cs.CL] <https://arxiv.org/abs/2405.14831>
- [13] Haoyu Han, Yu Wang, Harry Shomer, Kai Guo, Jiayuan Ding, Yongjia Lei, Mahantesh Halappanavar, Ryan A. Rossi, Subhabrata Mukherjee, Xianfeng Tang, Qi He, Zhigang Hua, Bo Long, Tong Zhao, Neil Shah, Amin Javari, Yinglong Xia, and Jiliang Tang. 2025. Retrieval-Augmented Generation with Graphs (GraphRAG). arXiv:2501.00309 [cs.IR] <https://arxiv.org/abs/2501.00309>
- [14] Sebastian Hofstätter, Jiecao Chen, Karthik Raman, and Hamed Zamani. 2022. FiD-Light: Efficient and Effective Retrieval-Augmented Text Generation. arXiv:2209.14290 [cs.CL] <https://arxiv.org/abs/2209.14290>
- [15] Lei Huang, Weijiang Yu, Weitao Ma, Weihong Zhong, Zhangyin Feng, Haotian Wang, Qianglong Chen, Weihua Peng, Xiaocheng Feng, Bing Qin, and Ting Liu. 2025. A Survey on Hallucination in Large Language Models: Principles, Taxonomy, Challenges, and Open Questions. *ACM Trans. Inf. Syst.* 43, 2, Article 42 (Jan. 2025), 55 pages. doi:10.1145/3703155
- [16] infiniflow team. 2024. *GitHub - infiniflow/ragflow: RAGFlow is an open-source RAG (Retrieval-Augmented Generation) engine based on deep document understanding*. <https://github.com/infiniflow/ragflow>
- [17] Jiajie Jin, Yutao Zhu, Xinyu Yang, Chenghao Zhang, and Zhicheng Dou. 2024. Flashrag: A modular toolkit for efficient retrieval-augmented generation research. *arXiv preprint arXiv:2405.13576* (2024).
- [18] Satyapriya Krishna, Kalpesh Krishna, Anhad Mohananeey, Steven Schwarcz, Adam Stambler, Shyam Upadhyay, and Manaal Faruqui. 2025. Fact, Fetch, and Reason: A Unified Evaluation of Retrieval-Augmented Generation. arXiv:2409.12941 [cs.CL] <https://arxiv.org/abs/2409.12941>
- [19] Patrick Lewis, Ethan Perez, Aleksandra Piktus, Fabio Petroni, Vladimir Karpukhin, Naman Goyal, Heinrich Küttler, Mike Lewis, Wen tau Yih, Tim Rocktäschel, Sebastian Riedel, and Douwe Kiela. 2021. Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks. arXiv:2005.11401 [cs.CL] <https://arxiv.org/abs/2005.11401>
- [20] Xinze Li, Sen Mei, Zhenghao Liu, Yukun Yan, Shuo Wang, Shi Yu, Zheni Zeng, Hao Chen, Ge Yu, Zhiyuan Liu, et al. 2024. RAG-DDR: Optimizing Retrieval-Augmented Generation Using Differentiable Data Rewards. *arXiv preprint arXiv:2410.13509* (2024).
- [21] Shervin Minaee, Tomas Mikolov, Narjes Nikzad, Meysam Chenaghlu, Richard Socher, Xavier Amatriain, and Jianfeng Gao. 2024. Large Language Models: A Survey. arXiv:2402.06196 [cs.CL] <https://arxiv.org/abs/2402.06196>
- [22] Kurnia Muludi, Kaira Milani Fitriani, Joko Triloka, and Sutedi. 2024. Retrieval-Augmented Generation Approach: Question Answering using Large Language Model. *International Journal of Advanced Computer Science and Applications* 15, 3 (2024). doi:10.14569/IJACSA.2024.0150379
- [23] neuml team. 2024. *GitHub - neuml/txtai: All-in-one open-source embeddings database for semantic search, LLM orchestration and language model workflows*. <https://github.com/neuml/txtai>
- [24] OpenAI, Josh Achiam, Steven Adler, Sandhini Agarwal, Lama Ahmad, Ilge Akkaya, Florencia Leoni Aleman, Diogo Almeida, Janko Altmenschmidt, Sam Altman, Shyamal Anadkat, et al. 2024. GPT-4 Technical Report. arXiv:2303.08774 [cs.CL] <https://arxiv.org/abs/2303.08774>
- [25] Jie Ouyang, Yucong Luo, Mingyue Cheng, Daoyu Wang, Shuo Yu, Qi Liu, and Enhong Chen. 2024. Revisiting the Solution of Meta KDD Cup 2024: CRAG. arXiv:2409.15337 [cs.IR] <https://arxiv.org/abs/2409.15337>
- [26] Bhaskarjit Sarmah, Benika Hall, Rohan Rao, Sunil Patel, Stefano Pasquali, and Dhagash Mehta. 2024. HybridRAG: Integrating Knowledge Graphs and Vector Retrieval Augmented Generation for Efficient Information Extraction. arXiv:2408.04948 [cs.CL] <https://arxiv.org/abs/2408.04948>
- [27] Parth Sarthi, Salman Abdullah, Aditi Tuli, Shubh Khanna, Anna Goldie, and Christopher D. Manning. 2024. RAPTOR: Recursive Abstractive Processing for Tree-Organized Retrieval. arXiv:2401.18059 [cs.CL] <https://arxiv.org/abs/2401.18059>
- [28] Timo Schick, Jane Dwivedi-Yu, Roberto Dessi, Roberta Raileanu, Maria Lomeli, Luke Zettlemoyer, Nicola Cancedda, and Thomas Scialom. 2023. Toolformer: Language Models Can Teach Themselves to Use Tools. arXiv:2302.04761 [cs.CL] <https://arxiv.org/abs/2302.04761>
- [29] SciPhi team. 2024. *GitHub - SciPhi-AI/R2R: The most advanced AI retrieval system. Agentic Retrieval-Augmented Generation (RAG) with a RESTful API*. <https://github.com/SciPhi-AI/R2R>
- [30] Yuchen Shao, Yuheng Huang, Jiawei Shen, Lei Ma, Ting Su, and Chengcheng Wan. 2025. Are LLMs Correctly Integrated into Software Systems? arXiv:2407.05138 [cs.SE] <https://arxiv.org/abs/2407.05138>
- [31] Aditi Singh, Abul Ehtesham, Saket Kumar, and Tala Talaie Khoei. 2025. Agentic Retrieval-Augmented Generation: A Survey on Agentic RAG. arXiv:2501.09136 [cs.AI] <https://arxiv.org/abs/2501.09136>
- [32] Jiejun Tan, Zhicheng Dou, Wen Wang, Mang Wang, Weipeng Chen, and Ji-Rong Wen. 2024. Htmrlag: Htm is better than plain text for modeling retrieved knowledge in rag systems. *arXiv preprint arXiv:2411.02959* (2024).
- [33] Ruobing Wang, Daren Zha, Shi Yu, Qingfei Zhao, Yuxuan Chen, Yixuan Wang, Shuo Wang, Yukun Yan, Zhenghao Liu, Xu Han, et al. 2024. Retriever-and-Memory: Towards Adaptive Note-Enhanced Retrieval-Augmented Generation. *arXiv preprint arXiv:2410.08821* (2024).
- [34] Xiaohua Wang, Zhenghua Wang, Xuan Gao, Feiran Zhang, Yixin Wu, Zhibo Xu, Tianyuan Shi, Zhengyuan Wang, Shizheng Li, Qi Qian, Ruicheng Yin, Changze Lv, Xiaoqing Zheng, and Xuanjing Huang. 2024. Searching for Best Practices in Retrieval-Augmented Generation. arXiv:2407.01219 [cs.CL] <https://arxiv.org/abs/2407.01219>
- [35] Zheng Wang, Shu Xian Teo, Jieer Ouyang, Yongjun Xu, and Wei Shi. 2024. M-RAG: Reinforcing Large Language Model Performance through Retrieval-Augmented Generation with Multiple Partitions. arXiv:2405.16420 [cs.CL] <https://arxiv.org/abs/2405.16420>
- [36] Diji Yang, Jinneng Rao, Kezhen Chen, Xiaoyuan Guo, Yawen Zhang, Jie Yang, and Yi Zhang. 2024. IM-RAG: Multi-Round Retrieval-Augmented Generation Through Learning Inner Monologues. arXiv:2405.13021 [cs.CL] <https://arxiv.org/abs/2405.13021>
- [37] Shi Yu, Chaoyue Tang, Bokai Xu, Junbo Cui, Junhao Ran, Yukun Yan, Zhenghao Liu, Shuo Wang, Xu Han, Zhiyuan Liu, et al. 2024. Visrag: Vision-based retrieval-augmented generation on multi-modality documents. *arXiv preprint arXiv:2410.10594* (2024).
- [38] Tan Yu, Anbang Xu, and Rama Akkiraju. 2024. In Defense of RAG in the Era of Long-Context Language Models. arXiv:2409.01666 [cs.CL] <https://arxiv.org/abs/2409.01666>

- [39] Zheni Zeng, Yuxuan Chen, Shi Yu, Yukun Yan, Zhenghao Liu, Shuo Wang, Xu Han, Zhiyuan Liu, and Maosong Sun. 2024. KBAIAlign: KBAIAlign: Efficient Self Adaptation on Specific Knowledge Bases. *arXiv preprint arXiv:2411.14790* (2024).
- [40] Jintian Zhang, Xin Xu, Ningyu Zhang, Ruibo Liu, Bryan Hooi, and Shumin Deng. 2024. Exploring Collaboration Mechanisms for LLM Agents: A Social Psychology View. *arXiv:2310.02124 [cs.CL]* <https://arxiv.org/abs/2310.02124>
- [41] Wen Zhang, Long Jin, Yushan Zhu, Jiaoyan Chen, Zhiwei Huang, Junjie Wang, Yin Hua, Lei Liang, and Huajun Chen. 2024. TrustUQA: A Trustful Framework for Unified Structured Data Question Answering. *arXiv:2406.18916 [cs.CL]* <https://arxiv.org/abs/2406.18916>
- [42] Penghao Zhao, Hailin Zhang, Qinhan Yu, Zhengren Wang, Yunteng Geng, Fangcheng Fu, Ling Yang, Wentao Zhang, Jie Jiang, and Bin Cui. 2024. Retrieval-Augmented Generation for AI-Generated Content: A Survey. *arXiv:2402.19473 [cs.CV]* <https://arxiv.org/abs/2402.19473>
- [43] Qingfei Zhao, Ruobing Wang, Yukuo Cen, Daren Zha, Shicheng Tan, Yuxiao Dong, and Jie Tang. 2024. LongRAG: A Dual-Perspective Retrieval-Augmented Generation Paradigm for Long-Context Question Answering. *arXiv:2410.18050 [cs.CL]* <https://arxiv.org/abs/2410.18050>
- [44] Wayne Xin Zhao, Kun Zhou, Junyi Li, Tianyi Tang, Xiaolei Wang, Yupeng Hou, Yingqian Min, Beichen Zhang, Junjie Zhang, Zican Dong, Yifan Du, Chen Yang, Yushuo Chen, Zhipeng Chen, Jinhao Jiang, Ruiyang Ren, Yifan Li, Xinyu Tang, Zikang Liu, Peiyu Liu, Jian-Yun Nie, and Ji-Rong Wen. 2024. A Survey of Large Language Models. *arXiv:2303.18223 [cs.CL]* <https://arxiv.org/abs/2303.18223>
- [45] Huichi Zhou, Kin-Hei Lee, Zhonghao Zhan, Yue Chen, Zhenhao Li, Zhaoyang Wang, Hamed Haddadi, and Emine Yilmaz. 2025. TrustRAG: Enhancing Robustness and Trustworthiness in RAG. *arXiv:2501.00879 [cs.CL]* <https://arxiv.org/abs/2501.00879>
- [46] Kunlun Zhu, Yifan Luo, Dingling Xu, Ruobing Wang, Shi Yu, Shuo Wang, Yukun Yan, Zhenghao Liu, Xu Han, Zhiyuan Liu, et al. 2024. Rageval: Scenario specific rag evaluation dataset generation framework. *arXiv preprint arXiv:2408.01262* (2024).

A Information Amplification Specialist Agent

To address the issue that high semantic similarity often leads to interference among retrieval results and significantly reduces the accuracy of question-answering systems, the Information Amplification Specialist analyzes the semantic features of each segmented block, identifies subtle differences, and enhances these distinctions by rephrasing the information. The specific prompt design is as follows:

You are an expert in system operations and manual interpretation, skilled in converting abstract operational steps into specific, vivid usage scenarios, helping users better understand and master system workflows. Based on a part of the operation manual and the current directory structure, you are to generate a unique usage scenario that simulates the actual system operation scenario.

The document is titled: "{title}", and the directory structure is: "{struct}"

The document content is as follows: "{context}"

The reference formula for the generated scenario is: In the XX platform, performing XX operation to complete XX task or achieve XX result. It satisfies: User_Scenario = R + S + O + R, where: 1. U represents the user role 2. S represents the platform 3. O represents the operation the user needs to perform (e.g., dragging a file) 4. R represents the function realized by the operation (e.g., file upload)

Only output the unique usage scenario you generate for this content.

First, understand the document content and summarize it. Then, supplement information based on the document's

directory structure, detailing the directory structure information, and generate according to the scenario reference formula.

B Interaction Analyst Agent

B.1 Our Approach

The Interaction Analyst refines the semantic information of the user's question, captures the user's intent, and reconstructs the question. The specific prompt design is as follows:

"Historical dialogue between user and system includes: historical dialogue information Current query: current query Requirements: First, please determine whether the current question conforms to the following rule: Q=W+R+S+O, where Q represents the user's question, which should conform to W, namely When, Where, How, Why; R represents the user's role; S represents the system where the user wants to inquire about operations; and O corresponds to the operation the user wants to query. If the current question has missing information, prioritize considering recent dialogue history. Infer the key information missing in the user's current query, and complete the query based on the context, making it more coherent and accurate semantically. Specific requirements are as follows:

Combine the historical dialogue to infer the background of the user's current query and the potential information not expressed; When completing the query, consider the user's intention and every aspect in the context to ensure the completed content is natural and logical."

B.2 Explicit Instruction

"" Please strictly adhere to the following when answering: 1. Prioritize analyzing dialogue content directly relevant to the user's most recent question 2. When historical information conflicts with the latest input, prioritize the latest input 3. Only reference earlier history when necessary

[Dialogue History] (most recent to earliest) history

[Current Query] current query

Please strictly follow the above principles to complete the user's question: ""

B.3 Structured Context

"" # Dialogue Flow Analysis Latest Interaction (must be prioritized): User: current_query

Relevant History : format_history(history)

Response Requirements 1. First, parse the core request of the latest interaction and determine if the user's question is clear 2. Only retrieve relevant history when necessary 3. Response must naturally connect with the latest interaction Please generate a response that reconstructs the user's question according to the structure above: ""

C GPT4o-score

C.1 The Design of Prompt

The specific criteria for scoring using GPT4o and the prompt design are as follows.

Evaluation Task Instruction You are a professional content quality assessment expert. Please strictly evaluate the quality of the model's response based on the following criteria.

Input Data [User Question] Insert the specific user query here

[Gold Standard Answer] Insert the standard reference answer here

[Response to be Evaluated] Insert the model response to be evaluated here

Evaluation Dimensions (Total: 100 points) 1. Factual Accuracy (40 points) - Complete match with core facts in gold standard answer: 36-40 points - Main facts correct but with minor deviations: 30-35 points - Contains critical factual errors: ≤ 29 points

2. Information Completeness (30 points) - Formula: Score = $30 \times (\text{number of key points covered} / \text{total number of key points in gold standard answer})$ - Additional points for effective supplementary information (3 points)

3. Logical Coherence (20 points) - Complete reasoning chain without gaps: 18-20 points - Contains repairable logical jumps: 14-17 points - Logical confusion affecting comprehension: 13 points

4. Expression Clarity (10 points) - Accurate expression and easy to understand: 9-10 points - Expression issues that do not affect understanding: 7-8 points - Requires repeated interpretation to understand: 6 points

Output Requirements :Json

C.2 Scoring Formula

$$GPT4o\text{-score} = 0.4 \times$$

$$\begin{cases} 40 & \text{if match score} = 1 \text{ (perfect match)} \\ 30 + 5 \times \text{match score} & \text{if } 0.75 \leq \text{match score} < 1 \text{ (major facts correct)} \\ \text{match score} \times 30 & \text{if } 0 < \text{match score} < 0.75 \text{ (key factual errors)} \end{cases}$$

$$+ 0.3 \times \left(30 \times \left(\frac{K_{\text{covered}}}{K_{\text{total}}} \right) + \Delta \right)$$

$$+ 0.2 \times \begin{cases} 20 & \text{if } L_{\text{logic}} = 1 \text{ (complete reasoning chain)} \\ 15 + 5 \times L_{\text{logic}} & \text{if } 0.5 \leq L_{\text{logic}} < 1 \text{ (logical gaps fixable)} \\ L_{\text{logic}} \times 15 & \text{if } L_{\text{logic}} < 0.5 \text{ (logical confusion)} \end{cases}$$

$$+ 0.1 \times \begin{cases} 10 & \text{if } C_{\text{clarity}} = 1 \text{ (clear and easy to understand)} \\ 8 + 2 \times C_{\text{clarity}} & \text{if } 0.7 \leq C_{\text{clarity}} < 1 \text{ (minor clarity issues)} \\ C_{\text{clarity}} \times 8 & \text{if } C_{\text{clarity}} < 0.7 \text{ (difficult to understand)} \end{cases} \quad (8)$$